



SISMO SFTP FILE UPLOADER

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SftpClient Setup and Usage Guide - macOS

This guide explains how to prepare, configure, and run SftpClient.jar on macOS. It also includes instructions for scheduling the application to run daily at 05:00 AM and for removing the scheduled task.

Note: 05:00 AM is configured in the local time zone. It is recommended to adjust the schedule so that file uploads occur at 5:00 AM UTC. **Note:** SftpClient is a Java-based application that connects to an SFTP server and uploads files to a specified directory.

Prerequisites

1. **Java Runtime Environment (JRE)**

Ensure the jre folder is included in the same directory as the SftpClient.jar.

2. **Environment Variables**

The run_macos.command script requires several environment variables to be set. Update these to match your environment before execution.

3. **Required Folders**

The following folders in the root directory of the application (created during the zip packaging) are used by default for file uploads:

- portfolio: for portfolio files.
- user_indicator: for stock-related indicator files.
- macro_indicator: for macro indicator files.

You may create your own folders and update the environment variables in run_macos.command accordingly.

Environment Variables

Below there are key environment variables, their purposes, and usage examples:

SFTP_SERVER

- **Purpose:** Specifies the hostname or IP address of the SFTP server to which files will be uploaded.
- **Example:** sftp.example.com or 192.168.1.100

SFTP_USER

- **Purpose:** The username used for authenticating with the SFTP server.
 - **Example:** sftpuser
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SFTP_PRIVATE_KEY

- **Purpose:** Path to the private SSH key file used for SFTP authentication.
- **Example:** /Users/youruser/.ssh/id_rsa

SFTP_PHRASE

- **Purpose:** Passphrase for the private SSH key, if required. Leave empty if the key is not encrypted.
- **Example:** yourSecretPassphrase
(Leave blank if not needed)

SFTP_LOCAL_USER_INDICATOR_DIRECTORY

- **Purpose:** Directory path containing user indicator files to be uploaded.
- **Example:** /Users/youruser/sftpclient/user_indicator

SFTP_LOCAL_MACRO_INDICATOR_DIRECTORY

- **Purpose:** Directory path containing macro indicator files to be uploaded.
- **Example:** /Users/youruser/sftpclient/macro_indicator

SFTP_LOCAL_PORTFOLIO_DIRECTORY

- **Purpose:** Directory path containing portfolio files to be uploaded.
- **Example:** /Users/youruser/sftpclient/portfolio

SFTP_PORTFOLIO_FILE_MAPPER

- **Purpose:** Maps external IDs to substrings in file names for portfolios, which are stored on a user's side, enabling automatic renaming before upload. There should be no hyphens/dashes in the portfolio names (e.g. Portfolio1 as below)
- **Example:** *PortfolioExternalId1-Portfolio1, PortfolioExternalId2-Portfolio2*
- **Mock File Name:** Portfolio1_Report.csv renamed to PortfolioExternalId1-[timestamp]-[operation].csv (The substring Portfolio1 in the file name matches the mapping PortfolioExternalId1-Portfolio1. The file is renamed to PortfolioExternalId1-[timestamp]-[operation].csv before being uploaded)

SFTP_PORTFOLIO_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Specifies the default operation for portfolio files.
 - F = replace file and delete history
 - M = modify file
- **Example:** M
Files will be uploaded with the “modify” (update) operation unless specified otherwise.

SFTP_USER_INDICATOR_FILE_MAPPER

- **Purpose:** Same logic as for portfolios, but applies to user indicator files.
- **Example:** *UserExternalId1-UserIndicator1, UserExternalId2-UserIndicator2*



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- **Mock File Name:** UserIndicator1_Data.csv renamed to UserExternalId1-[timestamp]-[operation].csv and zipped before upload

SFTP_USER_INDICATOR_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Default operation for user indicator files.
 - F = replace file and delete history
 - M = modify file
- **Example:** M
Files will be uploaded with the “modify” operation unless specified otherwise.

SFTP_MACRO_INDICATOR_FILE_MAPPER

- **Purpose:** Same mapping logic, applied to macro indicator files.
- **Example:** *MacroExternalId1-MacroIndicator1, MacroExternalId2-MacroIndicator2*
- **Mock File Name:** MacroIndicator1_Stats.csv → renamed to MacroExternalId1-[timestamp]-[operation].csv

SFTP_MACRO_INDICATOR_FILE_MAPPER_DEFAULT_OPERATION

- **Purpose:** Default operation for macro indicator files. F means replace the file, and M means modify the file.
- **Example:** F
Files will be uploaded with the “replace” operation unless specified otherwise.

✅ *Ensure all environment variables are correctly set in run_macos.bat before running the application.* If you encounter issues, check the log file (log.txt) in the application’s directory for details.

Manual Execution Steps

1. **Edit the Script**
2. **Make the File Executable:**
 - Open the terminal and navigate to the directory containing run_macos.command.
 - Run the following command to ensure it is executable:

```
chmod +x run_macos.command
```

3. **Run the Script**
Double-click the run_macos.command file in Finder or run command in the terminal:

```
chmod +x run_macos.command
`bash
./run_macos.command
```

4. **Verify Execution**
The script checks for the Java runtime and executes SftpClient.jar.



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5. Check Logs:

- The script writes logs to /tmp/sftpclient_debug.log.
- Open the log file to verify the execution or debug any issues:

```
cat /tmp/sftpclient_debug.log
```

Steps to Run SftpClient.jar Manually using Terminal

1. Adjust Environment Variables:

- Open run_macos.command in a text editor.
- Update the environment variables with the correct values for your environment.

2. Run the Script:

- Make the script executable:

```
chmod +x run_macos.command
```

- Execute the script:

```
./run_macos.command
```

3. Verify Execution:

- The script will check for the presence of the Java runtime and execute SftpClient.jar.

Schedule SftpClient.jar to Run Daily at 05:00 AM

1. Prepare the Script:

- Ensure schedule_macos.command is in the same directory as run_macos.command.

2. Run the Scheduler Script:

- Execute schedule_macos.command to create a launchd task:

```
./schedule_macos.command
```

3. Verify the Task:

- Check if the task is loaded:

```
launchctl list | grep com.user.sftpclient
```

Remove the Scheduled Task

1. Run the Removal Script:

- Execute remove_schedule_macos.command to remove the launchd task:

```
./remove_schedule_macos.command
```



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2. Verify Removal:

- Check if the task is unloaded:
`launchctl list | grep com.user.sftpclient`

Important Notes



- All .command scripts (run_macos.command, schedule_macos.command, remove_schedule_macos.command) must be in the same directory as SftpClient.jar and the jre folder.
- The folders user_indicator, macro_indicator, and portfolio are created during packaging. You may use custom folders by updating run_macos.command.
- If you encounter errors, check logs in the Task Scheduler or in log.txt.

File Structure

The directory should have the following structure:

```
/your-directory
├── run_macos.command
├── schedule_macos.command
├── remove_schedule_macos.command
├── SftpClient.jar
├── README_MACOS.md
├── README_MACOS.pdf
├── jre/
├── user_indicator/
├── macro_indicator/
└── portfolio/
```

Troubleshooting

- **Java Runtime Not Found**
Ensure the jre folder is present.
- **Task Scheduler Errors**
Confirm you have the required permissions. Check the History tab for details.
- **Incorrect Environment Variables**
Double-check the variables in run_macos.command for typos or invalid values.